

§3.3

Measures of Relative Standing

Where does one value sit inside the data?

Today's Goal

- Calculate the percentile rank of a data value
- Find a specific percentile from a dataset
- Calculate and interpret z-scores
- Use z-scores to compare values from different datasets

What You Will Be Graded On

LT.6 Numerical Summaries of Data

I understand the different numerical summaries for data (e.g., mean, median, mode, standard deviation, etc.). I can find and interpret the values of these numerical summaries for a given data set.

§3.1 and §3.2 described the whole dataset. Today we place one value inside it.

The Reservoir Is at 80% Full

A news report on a Central Valley reservoir, in late summer.

Should the city be worried?

You cannot tell from 80% alone. You need the other years.

A measure of relative standing is a number that reports the location of one data value against the rest of the same dataset.

What a Percentile Is

Definition 3.3.1

A **percentile** is a number indicating the value below which a given percentage of observations fall. The k -th percentile of a data set, denoted P_k , is a value such that approximately $k\%$ of the data lies below it.

There are essentially two types of questions that involve percentiles:

1. What is the percentile rank of a particular data value?
2. What data value corresponds to a particular percentile?

Counting What Is Below

Procedure 3.3.2 Finding a Percentile Rank

Given a value x from a dataset, we find its percentile ranking as follows:

1. Sort the data from smallest to largest.
2. Let a denote the number of data values that are less than x .
3. Calculate $\frac{a}{n} \cdot 100\%$, where n is the size of the dataset, and round to the nearest whole number.

Step 2 says *less than*, not *less than or equal to*.

Reading a Rank Off a Frequency Table

Forty students in a summer reading program reported how many books they read.

Books read	2	3	4	5	6	7
Students	3	6	9	13	7	2

Find the percentile rank of a student who read 5 books.

Add the frequencies: $n = 3 + 6 + 9 + 13 + 7 + 2 = 40$.

Fewer than 5 books: $3 + 6 + 9 = 18$ students, so $a = 18$.

$\frac{18}{40} \cdot 100\% = 45\%$, so a student who read 5 books is at the 45th percentile.

What Happens at the Top of the Data

Books read	2	3	4	5	6	7
Students	3	6	9	13	7	2

Now find the percentile rank of a student who read 7 books, the largest value in the table.

Fewer than 7 books: $40 - 2 = 38$ students, so $a = 38$.

$$\frac{38}{40} \cdot 100\% = 95\%.$$

The top readers land at 95%, not 100%. No data value has the whole dataset below it.

Back to the Reservoir

End-of-summer capacity, in percent, for the last ten years.

28 41 55 62 70 80 84 88 91 96

Is 80% full the 80th percentile?

Five years came in below 80, so $\frac{5}{10} \cdot 100\% = 50\%$.

An 80% year sits at the 50th percentile: half the ten years came in lower. The 80 says how full the reservoir got; the 50 says how that year compares to the rest.

From a Percentile to a Value

Procedure 3.3.4 Finding a Percentile

Given a dataset, we find the k -th percentile P_k as follows:

1. Sort the data from smallest to largest.
2. Calculate the locator position:

$$L_k = \frac{k}{100} \cdot (n + 1).$$

- 3a. If L_k is a whole number, then P_k is the value of the data set that appears in the L_k -th position.

When the Locator Lands Between

Procedure 3.3.4, continued

3b. If L_k is not a whole number, then we round L_k down and up. Call these values A and B . We know that P_k is somewhere between the two data values that appear in the A -th and B -th positions. Take the decimal part of L_k and multiply by the distance between these two data values. Then add this to the A -th data value to get P_k .

$$P_k = (A\text{-th value}) + (\text{decimal part of } L_k) \cdot [(B\text{-th value}) - (A\text{-th value})].$$

Running the Locator Formula

Daily high temperatures in degrees Fahrenheit, 24 days, sorted.

61	62	64	64	65	67	68	70	71	71	72	73
74	75	76	77	78	78	79	81	84	86	88	92

Find the 60th percentile.

$$L_{60} = \frac{60}{100} \cdot (24 + 1) = 0.60 \cdot 25 = 15.$$

Whole number, so P_{60} is the 15th value, which is 76 degrees.

The Same Data, a Different Percentile

61 62 64 64 65 67 68 70 71 71 72 73
74 75 76 77 78 78 79 81 84 86 88 92

Find the 30th percentile for the same 24 days.

$$L_{30} = \frac{30}{100} \cdot 25 = 7.5.$$

Not whole, so $A = 7$, $B = 8$: values 68 and 70, a distance of 2.

So $P_{30} = 68 + 0.5 \cdot (2) = 69$ degrees.

Let Excel Find the Percentile

Procedure 3.3.6 Finding a Percentile Using Excel

Excel provides us with the function `=PERCENTILE.EXC ( )`, which can be used to find the percentile as described in Procedure 3.3.4. You will need to provide it with two things:

1. where your data is located and
2. which percentile you are looking for (written in decimal form).

So if we entered the data from Example 3.3.5 into Column A of an Excel sheet, we would enter `=PERCENTILE.EXC ( A:A, 0.82 )` into any blank cell to compute P_{82} .

Not Every Method Agrees

Warning 3.3.7

There are quite a few different methods for finding percentiles and quartiles, not all of which produce the same results. Generally speaking, the difference among these methods is mostly only noticeable for small datasets.

Excel also offers `=PERCENTILE.INC ( )`. It uses a different rule and will not always match. This course uses `=PERCENTILE.EXC ( )`.

One Took the SAT, One Took the ACT

Test	Mean	Standard deviation	This student scored
SAT	1050	200	1310
ACT	20.5	5	28

Who did better on
their own test?

You cannot compare 1310 to 28 directly. Different scales.

Counting in Standard Deviations

Definition 3.3.8

The **z -score** corresponding to a particular data value x is the number z that tells us how many standard deviations x lies above or below the mean of the dataset:

$$\text{Sample: } z = \frac{x - \bar{x}}{s}, \quad \text{Population: } z = \frac{x - \mu}{\sigma}.$$

The z -score is generally rounded to two decimal places. The inverse formulas recover x from a known z :

$$\text{Sample: } x = \bar{x} + z \cdot s, \quad \text{Population: } x = \mu + z \cdot \sigma.$$

What the Sign and the Size Tell You

- $z > 0$: the value is above the mean.
- $z < 0$: the value is below the mean.
- $z = 0$: the value is the mean.

The size is the distance, measured in standard deviations. A z of 2 means two standard deviations away from the mean.

A value with $|z| > 2$ is considered unusual.

A z -score carries no units, which is what lets us compare values from different datasets.

Forward: From a Value to a z-Score

Procedure 3.3.9 Computing a z-Score (and its Inverse)

Forward (given x , find z):

1. Compute the mean ($\bar{x}$ or μ) and standard deviation (s or σ).
2. Compute the z -score as

$$z = \frac{x - \bar{x}}{s} \quad (\text{sample}), \quad z = \frac{x - \mu}{\sigma} \quad (\text{population}).$$

3. Usually rounded to two decimal places, but not always. (More about this convention later.)

Inverse: From a z-Score to a Value

Procedure 3.3.9, continued

Inverse (given z , find x):

1. Compute $\bar{x}$ and s (or μ and σ) as above.
2. Compute the data value as

$$x = \bar{x} + z \cdot s \quad (\text{sample}), \quad x = \mu + z \cdot \sigma \quad (\text{population}).$$

Keep four decimals in $\bar{x}$ and s . Rounding them early moves the second decimal of z .

A z-Score From Real Data

Same 24 temperatures. In Excel, =AVERAGE (A : A) and =STDEV . S (A : A) give

$$\bar{x} = 74, \quad s \approx 8.3458.$$

Find the z -score of a day whose high was 88 degrees.

$$z = \frac{x - \bar{x}}{s} \approx \frac{88 - 74}{8.3458} \approx 1.68.$$

About 1.68 standard deviations above the mean. Under 2, so not unusual.

Working Backward to the Value

Same sample, same summary numbers:

$$\bar{x} = 74, \quad s \approx 8.3458.$$

Find the temperature whose z -score is -1.5 .

$$x = \bar{x} + z \cdot s \approx 74 + (-1.5)(8.3458) \approx 61.5.$$

About 61.5 degrees, below the mean, exactly as the negative sign promised.

So Who Did Better?

Test	Mean	Standard deviation	This student scored
SAT	1050	200	1310
ACT	20.5	5	28

$$z_{\text{SAT}} = \frac{1310 - 1050}{200} = 1.30, \quad z_{\text{ACT}} = \frac{28 - 20.5}{5} = 1.50.$$

The ACT score is stronger: 1.50 beats 1.30.